PRACTICAL NO 4: FACE DETECTION

```
import cv2

# Load the pre-trained Haar Cascade face detector
face_cascade = cv2.CascadeClassifier(cv2.data.haarcascades + 'haarcascade_frontalface_default.xml')


# Load the image
image = cv2.imread("D:\CV\IMG1.jpg")


# Convert the image to grayscale (face detection works on grayscale images)
gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)


# Detect faces in the image
faces = face_cascade.detectMultiScale(gray, scaleFactor=1.1, minNeighbors=5, minSize=(30, 30))


# Draw rectangles around the faces
for (x, y, w, h) in faces:
    cv2.rectangle(image, (x, y), (x+w, y+h), (0, 255, 0), 2)


# Display the result
cv2.imshow('Face Detection', image)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

Output :

